INCH-POUND

MIL-M-70742 (AR)
30 September 1988

MILITARY SPECIFICATION

MOUNT, TELESCOPE AND QUADRANT: P\N 12599166

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers performance requirements, quality assurance provisions and packaging data relative to a telescope mount designed primarily to provide adjustments in pitch

2. APPLICABLE DOCUMENTS

2.1 The following documents~ of the issue in effect on date. of invitation for bids or request for proposals, form a part of this specifications to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Material; General Specification Governing the Manufacture and Inspection
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

U.S. Army, Armament Research, Development and Engineering Center (ARDEC)

5549108	Collimator
8565556	Pressure Tester
8658940	Gunners Quadrant, Calibrated (Inspection Aid)
11730831	Test Fixture
12599166	Mount, Telescope and Quadrant: P/N12599166

PACKAGING DATA SHEET

12599166	Packaging of Mount Telescope and Quadrant: P\N 12599166

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DSAM 4145.8	Radioactive quantities in the DOD supply system.

DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL
49 CFR 171-190 - Interstate Commerce Commission and Department of Transportation Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402).

3. REQUIREMENTS

3.1. Radioactive material. Prospective bidders should cognizant of the need for an Nuclear Regulatory Commission License governing the procurement, processing, handling and shipping of radioactive material.

3.1.1 Regulations pertaining to radioactive devices. Marking, identification, handling and shipping of radioactive devices shall be in accordance with MIL-STD-129 and DSAM 4145.8 as applicable.

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3.2 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Fabrication. The Mount shall be manufactured in accordance with Drawing F12599166 and drawings pertaining thereto.

3.4 General specification. The contractor shall be responsible for adherence to the requirements of MIL-F-13926.

3.5 Environmental.

3.5.1 Operating temperature. The mount shall meet the applicable requirements of 3.6.3 to 3.6.6.1 inclusive while exposed and thermally stabilized at ambient temperatures of +60 degrees Fahrenheit (F) to + 90 degrees Fahrenheit. The mount shall then meet the requirements of 3.6.5 at +150 degrees and -50 degrees Fahrenheit, subsequent to which the mount shall meet the requirements of 3.5.4.

3.5.2 Storage temperature. The mount shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature (+60 degrees to + 90 degrees Fahrenheit) after having been and thermally stabilized at ambient temperatures of +160 degrees F and -80 degrees F for a period of 6 hours at each temperature extreme.

3.5.3 Shock. The mount shall withstand a total of 18 shock impluses, three in each direction of three mutually perpendicular axis. Each shock impulse shall be a half-sine wave with a time duration of 3 $\pm$ 1 milliseconds.

3.5.4 Vibration.

3.5.4.1 Vibration “A”. The mount shall withstand a total of 270 minutes + 5 minutes of sweep-cycle vibration. The vibration shall be applied for 90 minutes $\pm$ 2 minutes along each of three mutually perpendicular major axis. A complete sweep-cycle shall consist of vibration from origin (5hz at 1 inch double amplitude) to mid point (5g’s $\pm$ 0.5 g's at 500 hz) to origin, and shall have a duration of 15 minutes $\pm$ minutes. Double amplitude shall be constant at 1 inch between 5 hz and 10 hz and varied with frequency to maintain a constant 5 g’s + 0.5 g’s acceleration between 10 hz and 500 hz. Upon completion of vibration the mount shall exhibit no evidence of damage.

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3.5.4.2 Vibration "B". The Mount shall be vibrated in a vertical plane at a constant frequency of 30 ± 5 cycles per seconds with an amplitude of 1/16 inch (1/8 ± 1/64 inch total excursion) for a period of five minutes plus or minus 15 seconds. Subsequent to vibration there shall be no evidence of physical failure.

3.5.5 External tritium contamination. Tritium shall not be present on the external surfaces of the mount in excess of 1000 dpm/100 sq. cm.

3.5.6 Sealing (counter box area). The counter box area of the mount shall be capable of sustaining an internal pressure of 2.0 ± .2 psi for a period of 2 hours (minimum) without loss of pressure in excess of 0.2 psi.

3.5.7 Purging. The counter box area of the mount shall be purged with dry nitrogen supplied at a pressure of 5 ± .2 psi until the dew point of the emergent gas is no higher than -25 degrees Fahrenheit. Purging apertures shall be quickly and positively closed to minimize intermingling of external atmosphere upon disconnecting of puring equipment.

3.5.8 Rain test. The mount shall meet the requirements of Procedure I of the rain test of MIL-F-13926.

3.6 Performance.

3.6.1 Orientation. The mount shall be positioned and orientated as follows:

a. The mounting surface (internal surface) of the mount shall be in a vertical plane.

b. The mount locating keyway shall be horizontal within 10 seconds.

The top locating surface for the telescope, shall be parallel to the mount's locating keyway within 0.5 mil.

d. The elevation and cross level vial bubble shall be centered.

e. The gunner's quadrant pads shall be level in reference to location keyway within 0.2 mil.

f. The locating key for the telescope shall be parallel to the vertical mounting surface within 1.0 mil.

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3.6.2 Movement.

3.6.2.1 Elevation travel. The mount shall travel through an excursion of at least 1333 mils in elevation and 89 mils in depression from the position established in 3.6.1 when the correction counter is set at zero.

3.6.2.2 Correction counter. The correction counter shall travel through an excursion of plus and minus 9 mils.

3.6.2.3 Cross level travel. The cross level knob shall move the mount through a minimum angle of 20 degrees to the right and 20 degrees to the left from the position established in 3.6.1.

3.6.3 Mount rigidity. With the mount positioned as in 3.6.1 apply a 20 pound load at the right side of the telescope mounting flange in line with the locating pins. Release the load and record the movement. Apply the same load at the left side of the telescope mounting flange in line with the locating pins. Release the load and record the movement. The total movement as indicated by the two readings shall not exceed the values of Table I.

TABLE I

MOVEMENT RIGIDITY

ELEVATION (in mils)	CANT (in degrees)	TOTAL MOVEMENT (in mils)
0	0	.25
800	0	.75
1100	0	1.75
1333	0	3.50

3.6.4 Accuracy. With the mount positioned in accordance with 3.6.1, the following conditions shall be met.

3.6.4.1 Plumb travel. With the cross level vial bubble centered, the line of sight shall not deviate from a plumb line by more than 0.5 mil when the elevation knob is actuated through its full range of movement.

3.6.4.2 Elevation error. Elevation error shall not exceed 0.5 mil excluding blacklash as read in 200 mil increments between zero and 1200mils.

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3.6.4.3 Backlash.

3.6.4.3.1 Elevation mechanism. Backlash in the elevation mechanism shall not exceed 0.5 mil.

3.6.4.3.2 Cross level mechanism. Backlash in the cross level mechanism shall not exceed 0.5 mil.

3.6.4.4 Correction counter. The correction counter shall be checked at plus and minus readings of 5 and 9 mils. The angle of the gunner quadrant seat shall correspond to the correction values within 0.2 mil when the elevation counter is reset to zero reading.

3.6.4.5 Counter alignment. Variation in the alignment of the numbers in the counter shall not exceed one sixteenth inch (1/16").

3.6.4.6 Correction counter operation. Plus values of 5 and 9 mils put on the correction counter shall subtract an equal number of mils from the elevation counter. Minus values of 5 and 9 mils put on the correction counter shall add an equal number of mils to the elevation counter.

3.6.5 Operation torques. The maximum of operating torques shall not exceed the values in Table II at temperatures indicated therein.

TABLE II

	AT STANDARD AMBIENT TEMPERATURE OF +60°F T0 +90°F AND AT +150° l		AT -50°F
Cross Level Knob	4 lb-inch min	12 lb-inch max	25 lb-inch max
Elevation Knob	4 lb-inch min	14 lb-inch max	25 lb-inch max
Correction Knob Elevation	2 lb-inch min	4 lb-inch max	8 lb-inch max

3.6.6 Correction.

3.6.6.1 Cant correction. The mount shall make left and right cant correction in accordance with the values in TABLE III. The elevation counter shall read within plus or minus 0.5 mils of the elevation angle of the quadrant seat.

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TABLE III

CANT CORRECTION

QUADRANT Elevation (in mils)	CANT ANGLE (in mils)	CORRECTION (in mils)	TOLERANCE (in mils)
500	**177.8**	96.1	**0.5**
800	177.8	**180.5**	**1.0**
1000	177.8	271.9	2.0

3.7 Illumination. The level vial bubble graduation lines and two counters shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into darkness with and without the use of a 50% neutral density filter.

3.8 Interchangeability. Interchangeability shall be verified by the insertion, seating, securing and subsequent removal of **maximum fit gage** on the mount's top surface.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specificaton. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval test shall consist of three (3) mounts plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the

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same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection, with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval test, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP'S the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and test related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Table IV herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables IV and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE IV - CLASSIFICATION OF DEFECTS

NO	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
MAJOR:	100% INSPECTION		
101	Sealing	3.5.6	4.6.1.6
102	Purging	3.5.7	4.6.1.7
103	Orientation	3.6.1	4.6.2.1
104	Illumination	3.7	4.6.3
105	Interchangeability	3.8	4.6.4
MAJOR :	AQL 0.65% DEFECTIVE		
106	Vibration “B”	3.5.4.2	4.6.1.4
107	External Tritium Contain	3.5.5	4.6.1.5
108	Elevation Travel	3.6.2.1	4.6.2.2
109	Correction Counter	3.6.2.2	4.6.2.3
110	Cross Level. Travel	3.6.2.3	4.6.2.4
111	Mount Rigidity	3.6.3	4.6.2.5
112	plumb Travel	3.6.4.1	4.6.2.6
113	Elevation Error	3.6.4.2	4.6.2.7
114	Elevation Backlash	3.6.4.3.1	4.6.2.7
115	Cross Level Backlash	3.6.4.3.2	4.6.2.4
116	Correction Counter	3.6.4.4	4.6.2.8
117	Counter Alignment	3.6.4.5	4.6.2.9
118	Correction Counter Oper	3.6.4.6	4.6.2.10
119	Operating Torques	3.6.5	4.6.2.11
120	Cant Correction	3.6.6.1	4.6.2.12
MINOR:	NONE		

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejections" as specified in MIL-STD-105.

4.4 Special. sampling.

4.4.1. General. Three telescopes shall be selected at random as a special sampling from each 100 produced or one months production whichever occurs first. The sample shall meet the requirements of Table V.

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TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.3	Applicable drawings - visual
302	General Specification	3.4	MIL-F-13926
303	Operating Temperature	3.5.1	4.6.1.1
304	Storage Temperature	3.5.2	4.6.1.1
305	Shock	3.5.3	4.6.1.2
306	Rain	3.5.8	4.6.1.8

4.4.2 Failure of sample. Should any one of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Test equipment.

4.5.1 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.3. Contractor furnished inspection equipment.

4.5.3.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the list of inspection documents when provided with the Technical Data Package (TDP).

4.5.3.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures” specified in this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials,

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dimensions and tolerance limit used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3.3 Test equipment. In conjunction with 4.5.3.3, the following standard test equipment shall be utilized in the performance of the applicable test as specified in 4.6.

	NOMENCLATURE	DESCRIPTION
1.	Hot & Cold Chamber	Standard type conforming to the accuracies specified under test facilities of MIL-F-13926
2.	Light source standard	
3.	Torque wrench	Standard shop type accurate to 10% of drawing tolerance

4.5.3.4 Device to test performance. Test equipment used to perform acceptance shall be equal to or exceed the accuracies depicted on drawing 11730831. Any equipment used to perform the tests shall be submitted to the government for review (see 6.3).

4.6 Examinations and tests.

4.6.1 Environmental.

4.6.1.1 Storage and operating temperatures. The mount shall meet the applicable requirements of 3.6.3 to 3.6.6.1 inclusive while exposed and thermally stabilized at ambient temperatures of + 60 degrees Fahrenheit (F) to + 90 degrees Fahrenheit. The mount shall then be placed in a government approved test chamber and the temperature stabilized at +150 degrees Fahrenheit and meet the requirement of 3.6.5. The temperature shall then be stabilized at +160 degrees Fahrenheit for a period of 6 hours. The mount shall then be thermally stabilized at a temperature of - 50 degrees Fahrenheit and meet the requirement of 3.6.5. The mount shall then be stabilized at an ambient temperature of - 80 degrees Fahreheit for a period of 6 hours. subsequent to the temperature cycle, the mount shall be visually and examined and shall show no evidence of physical failure and meet all requirements of this specification at standard ambient temperature (+60 degrees to +90 degrees Fahrenheit].

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4.6.1.2 Shock. The shock-producing apparatus shall be capable of producing the shock pulse and shall be calibrated for conformance with the requirement as outlined in 3.5.3. The frequency response of the complete measuring system, from any accelerometer through the readout instrument, shall be accurate to within ten percent of the specified tolerance. Subsequent to the 18 shock impulses the mount shall be visually examined for damage and shall meet all requirements of this specification.

4.6.1.3 Vibration "A". This test is applicable to the First Article sample only. The vibration exciter or transition table shall be approved by the government. The mount shall be attached by its normal means, either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration exciter table in order to minimize the effects of unbalanced loads. The input control sensing device(s) shall be rigidly attached to the vibration table or fixture if used, as near as possible to the attachment point of the mount. The vibration shall applied as specified in 3.5.4.1. Upon completion of the vibration test, the mount shall meet the requirements of this specification.

4.6.1.4 Vibration "B". This test is to be performed as outlined in Procedure V of MIL-F-13926, however the frequency and duration shall be as outlined in 3.5.4.2. At the conclusion of the test the mount shall meet the requirements of this specification.

4.6.1.5 External tritium contamination. Contamination test shall be performed by wiping the telescope with filter paper moistened in distilled water. The wiping shall be performed with moderate finger pressure. The damp filter paper shall be placed in a container with the proper portion of scintillation liquid. The read out shall be within 10% of the known standard value when counted to a total of 2000 disintergrations. The actual test for determining contamination is performed by placing the container into the scintillation system. The measured contamination shall be within the limits specified in 3.5.6.

4.6.1.6 Sealing (counter box area). The counter box portion of the mount shall be pressurized internally to the psi value specified in 3.5.6 with dry nitrogen for the length of time specified in 3.5.6. Dry nitrogen used in this test shall have a

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dew point of at least -25°F. Accuracy of the test equipment used on this test shall be equal to or exceed the accuracies depicted on Drawing F8565556.

4.6.1.7 Purging. The counter box area of the mount shall be flushed with dry nitrogen gas to conform to the requirements of 3.5.7. A dew point tester (standard shop type test equipment) shall be used to measure the nitrogen dew point for conformance with 3.5.7. Purging to be performed at standard ambient temperature and prior to performing temperature test. Pressure of nitrogen remaining within the counter box area to be between .25 and .5 psi.

4.6.1.8 Rain. The mount shall be subjected to the rain test outlined in Procedure I of MIL-F-13926. After completion of the rain test, a visual examination shall be made to determine leakage.

4.6.2 Performance testing.

4.6.2.1 Orientation. Using test equipment as described in 4.5.3.4, the mount shall be set up and checked for conformance with 3.6.1.

4.6.2.2 Elevation travel. With the mount and fixture Positioned as outlined in 4.6.2.1, rotate the mount's elevation knob to reflect maximum elevation. Note the reading on the elevation counter. Set this reading into Inspection Aid (Gunner's Quadrant) A8658940. Position the inspection aid on the quadrant seat assembly and by means of the fixture elevation handwheel, relevel the inspection aid. Observe the elevation level bubble of the mount with respect to the quadrant level vial. Both must indicate level within the limits outlined in 3.6.2.1. Repeat this procedure in the opposite direction. Any non-compatibility in level vial bubble position shall be measured by means of the quadrant.

4.6.2.3 Correction counter. Rotate the correction counter knob to indicate that the maximum correction range is within the limits of 3.4.2.2.

4.6.2.4 Cross level travel and backlash. With the mount positioned on the fixture as outlined in 4.6.2.1 and with the auxiliary telescope's sighting device assembled to the mount, rotate the cant mechanism to full extent of travel. Place the inspection aid over the appropriate set of pads of the sighting device and read the cant angle. This reading must be equal to or exceed the minimum requirement outlined in 3.6.2.3. Repeat this procedure at the opposite extreme setting. Return the cross leveling mechanism somewhere between zero and maximum angle and scribe a line on the casting and cross leveling knob and obtain a

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cross level angle reading using the inspection aid, record this reading. Continue to rotate the cross leveling knob one revolution beyond the index mark and return to the index mark. Exercise care not to override this point of coincidence and record this reading using the inspection aid. The difference, if any, is considered backlash and must conform to the limits outlined in 3.6.4.3.2.

4.6.2.5 Mount rigidity. With the fixture positioned as outlined in 4.5.2.3, position the mount as specified in 3.6.1 with the level vial bubbles centered. Check rigidity by gradually applying and releasing the pressure specified in 3.6.3 at the top left side of the telescope support as specified in 3.6.3. Take the reading on the test target and record the movement. Repeat this procedure gradually applying and releasing the specified pressure at the top right side of the telescope support. Take the reading on the test target and record the movement. The total movement shall be within the allowable maximum deviation specified in Table I of 3.6.3. The same procedure shall be followed as the simulated gun trunnion (cross leveling fixture) is moved from the "O" position to the 1333 mil elevation position to conform to the requirements of Table I. The maximum movement at the various settings shall not exceed the mil values specified in Table I.

4.6.2.6 Plumb travel. With this mount and fixture set as outlined in 4.6.2.1, this test is made by sighting through viewing telescope F5549108 and observing coincidence between the telescope's optical axis and the intersecting point of target centerlines while simultaneously rotating the mount in elevation and the test fixture an equal amount in the opposite direction. The optical line of sight shall not deviate from the intersecting point by more than the limits outlined in 3.6.4.1 as measured by target mil blocks. During this total travel the cross level vial shall remain level within the limits of 3.6.4.1.

4.6.2.7 Elevation error and backlash. With the fixture and mount positioned as outlined in 4.6.2.1, insert an elevation reading into the mount. Use the elevation counter as the recording mechanism. Insert this selected reading into the inspection aid and position the inspection aid on the quadrant seat assembly and by means of the fixture elevation handwheel relevel the quadrant level vial, do not override this level position. Observe coincidence between both level vial bubbles. Any non-compatibility between level vial bubble positions shall be measured by means of the quadrant: This difference, if any, shall not exceed the limits outlined in 3.6.4.2. Note this reading. Continue to rotate the elevation knob approximately one revolution beyond this point of coincidence. Return the knob to reposition the elevation level vial bubble to indicate level. Note the elevation counter reading. The difference, if any, in elevation reading is considered backlash and must be within the limits outlined in 3.6.4.3.1.

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4.6.2.8 Correction counter. With the mount and fixture set as outlined in 4.6.2.1, rotate the correction counter to the minimum setting specified in 3.6.4.4. Return the elevation counter to zero. By means of the fixture, level the elevation level vial of the mount. Do not override this level position. Place the inspection aid on the quadrant seat assembly and set the inspection aid vial to indicate level. Read the angle on the inspection aid. This reading to be identical to the correction counter reading within the limits of 3.6.4.4. Repeat this procedure at the setting specified in 3.6.4.4 each side of zero correction.

4.6.2.9 Counter alignment. Actuate the elevation knob through its complete travel range. During this travel make a minimum of 2 stops each side of zero elevation, and by standard measuring equipment? determine if digit alignment is within the limits outlined in 3.6.4.5.

4.6.2.10 Correction counter operation. This test may be performed in conjunction with the previous test. Visually determine that when a negative correction is inserted in the mount, the elevation counter reading increases an equal amount. Also, determine that when a positive correction is inseted in the mount, the elevation counter reading decreases an equal amount. Using the test procedure outlined in 4.6.2.1, determine that the inserted correction factors are recorded on the elevation counter as specified in 3.6.4.6.

4.6.2.11 Torques. All torque readings specified in 3.6.5 shall be measured using standard torque wrench and appropriate adapters, All readings to conform to the values specified in 3.6.5 for the temperature specified.

4.6.2.12 Cant operation. With the mount and fixture positioned as outlined in 4.6.2.1, rotate the auxiliary telescope’s line of sight 1600 mils and engage the locating pin in the proper locating hole indicated in the set up instructions similar to 117308731. The line of sight shall coincide with the vertical target line with the limits outlined, in 3.6.6.1. Elevation angle of the quadrant seat shall coincide with the elevation reading shown on the elevation counter within the limits of 3.6.6.1. Elevate and cant the fixture to selected set of values (Table III) and set the mount off an equal amount in the opposite direction. The telescope’s line of sight shall coincide with the target vertical centeline within the limits outlined in 3.6.6.1.

4.6.3 Illumination. Place 50% neutral density filter in front of the counters and level vials. In a darkened area visually inspect for conformance with 3.7.

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4.6.4 Interchangeability. Each mount shall be inspected for maximum fit conditions using the interchangeability gage 11747889. Any mount which-does not accept the interchangeability gage shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-P-14232, MIL-STD-129 and applicable packaging data sheets. The level of protection shall be specified in the procurement document.

6. NOTES

6.1 Intended use. The Telescope Mount and Quadrant: P/N 12599166 is primarily intended to support a panoramic telescope on the Light Howitzer, M119. The mount provides cant correction and elevation adjustments for the telescope.

6.2 Ordering data. Procurement documents should specify:

a. Title, date and number of this specification.

b. Serial numbers to be used for identification marking.

c. Applicable packaging data sheet number.

d. Selection of applicable levels of preservation and packaging.

e. Requirements for first article testing.

f. Contract data requirements for submission of inspection equipment designs to Data Item Description DI-R-1714.

6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection shall be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research Development and Engineering Center, ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, New Jersey 07806-5000. When the contractor submits inspection equipment designs to the government for approval, he shall give the following information in his letter of transmittal:

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a. The contract number.

b. The contract item (name, model number, etc.)

The designs remaining to be submitted and the expected date of submittal.

6.4 Subject term (key word) listing).

Mount
Telescope
Quadrant
Fire Control
Radiation

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240 -A926)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-70742 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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